CMU Capstone Checkpoint 4.1: Tree of Thought Reasoning

Project: Agentic Public Records Compliance Assistant

Selected Agent: Research Assistant

Is Tree of Thought Appropriate?

I have decided to use **Tree of Thought (ToT) reasoning selectively** in my Agentic Public Records Compliance Assistant.

The agent helps public-records staff examine documents, audio, and video requested under New Jersey's Open Public Records Act (OPRA). It identifies potentially sensitive information, retrieves relevant policies, recommends redactions, and provides supporting explanations for human review.

In previous checkpoints, I introduced ReAct reasoning, memory, external tools, and semantic retrieval using official NJ OPRA/GRC policies.

ReAct works well for straightforward cases where retrieved policies clearly support a recommendation. However, some records may involve multiple applicable policies, exceptions, or conflicting evidence. A linear reasoning approach may commit to the first matching policy without considering other possibilities.

ToT addresses this limitation by exploring multiple evidence-based interpretations before selecting a recommendation. I will activate ToT only when the initial policy evaluation identifies ambiguity or conflicting information.

Where ToT Will Be Used

ToT will operate **after sensitive-information detection and policy retrieval, but before the final redaction recommendation**.

For example, a synthetic police transcript may contain a personal phone number alongside information about an ongoing investigation. Multiple policies may apply, and the correct recommendation could depend on exceptions or additional facts.

Instead of immediately selecting one interpretation, the agent explores three possibilities:

- **Branch A:** Recommend redaction based on a supporting policy.
- **Branch B:** Investigate whether an exception or conflicting policy applies.

- **Branch C:** Escalate because the available evidence is insufficient.

This prevents premature commitment while ensuring that recommendations remain grounded in retrieved sources.

Proposed ToT Structure

Each **thought** represents a possible policy interpretation or next action supported by available evidence.

A **node** contains the detected information, retrieved policy references, assumptions, proposed action, and unresolved questions.

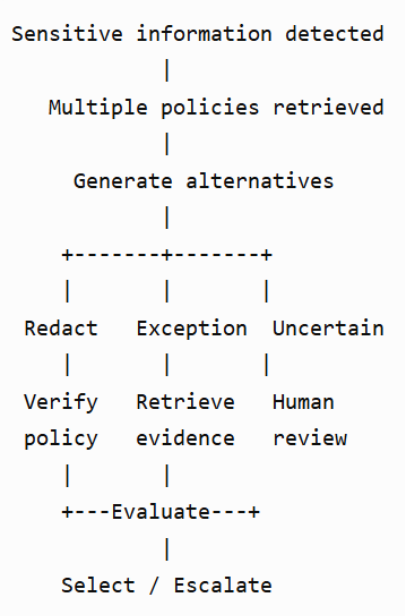
A **branch** represents an alternative interpretation or action. **Depth** indicates the number of successive reasoning or verification steps explored along that branch.

To keep the capstone manageable, I will use:

- **Branching factor:** Maximum three candidates per decision.
- **Maximum depth:** Three reasoning levels.
- **Beam width:** Retain the two strongest candidates at each level.
- **Termination:** Stop when sufficient verified evidence supports a recommendation, no useful branches remain, or the depth, time, or tool-call limit is reached.

If no interpretation has sufficient evidence, the agent escalates to human review.

An example reasoning tree:



Search and Evaluation Strategy

I will use **beam search**, which provides a balance between exploring alternatives and controlling computational cost.

Breadth-first search may generate too many branches, increasing processing time. Depth-first search may focus excessively on one interpretation. Monte Carlo-style sampling introduces additional variability that is unnecessary for a limited compliance workflow.

Beam search retains only the highest-ranked candidates at each reasoning level.

Each branch will be evaluated using a 100-point rubric:

Evaluation criteria	**Weight**
Relevance of retrieved policy evidence	30
Accuracy and verifiability of citations	30
Consistency with known facts and policy conditions	25
Explanation quality and uncertainty handling	15
Total	**100**

A critic agent will assess relevance and explanation quality, while deterministic checks verify citations, supporting evidence, and consistency with known facts.

Pruning rules:

- Branches containing fabricated citations or unsupported factual assumptions are rejected.
- Branches scoring below 70/100 are not eligible for a supported recommendation.
- Only the two highest-scoring eligible branches continue at each level.
- If scores tie, preference goes to the branch supported by stronger primary-source evidence.
- Unresolved material conflicts result in human review.

To control latency and cost, ToT will run only for ambiguous cases. Each execution will have a maximum depth of three, beam width of two, and a limited number of tool calls.

The score helps prioritize investigation; it does not independently establish legal correctness.

Mapping ToT Roles to Implementation Tools

I plan to use CrewAI, LangChain, and MCP with separate responsibilities.

ToT role	Tool	Responsibility
Thought Generator	CrewAI	Generates alternative evidence-based interpretations
Critic / Evaluator	CrewAI + LangChain	Evaluates candidate branches and applies validation checks
Decision Maker / Controller	LangChain	Coordinates beam search, pruning, and termination
Memory / State Manager	LangChain + application state	Maintains branch histories, evidence, scores, and workflow status
External Tool Access	MCP	Exposes policy search and evidence-validation tools

CrewAI provides specialized agent roles. LangChain supports tool integration and search orchestration. MCP provides a consistent interface for invoking external tools.

Branch state and decision history will be managed by the application rather than relying on MCP itself for persistent memory.

These components will reuse the existing policy retrieval and detection tools.

Updated Agent Workflow

The revised architecture will follow this sequence:

1. Receive a synthetic public-record request.
2. Use ReAct to examine the record and identify potentially sensitive information.
3. Retrieve relevant NJ OPRA/GRC policies through semantic search.
4. Evaluate whether the retrieved evidence is sufficient and unambiguous.
5. For straightforward cases, generate a policy-supported recommendation.
6. For ambiguous cases, invoke ToT to generate and evaluate alternative interpretations.
7. Prune weak or unsupported branches and select a supported outcome or escalate.

8. Present recommendations, evidence, and uncertainty to a human reviewer.
9. Record reviewer feedback and update the workflow state.

The agent will not independently authorize public release.

ToT Risk and Mitigation

A significant risk is **premature pruning**, where a valid branch receives a low score because important supporting information was not retrieved.

For example, an applicable exception may initially appear unsupported because the relevant policy passage is missing.

To manage this risk, branches with incomplete evidence will be marked as uncertain rather than immediately rejected. The controller may perform one additional targeted retrieval before pruning.

If the evidence remains insufficient, the agent will escalate the issue for human review rather than choose an unsupported interpretation.

Strict branching, depth, and tool-call limits will also prevent excessive processing.

Conclusion

My refined design combines **ReAct for routine evidence gathering and ToT for ambiguous policy decisions**.

ReAct keeps straightforward processing efficient, while ToT helps compare evidence-based alternatives when multiple interpretations may exist.

This approach preserves the original capstone's focus on reliability, traceability, and human oversight while introducing more advanced reasoning where it provides meaningful value.

The project will continue using publicly available policies and synthetic records. All final disclosure decisions will remain with a human reviewer.